

WeasyPrint: HTML/CSS to PDF

What is WeasyPrint?

WeasyPrint converts HTML and CSS to beautiful PDFs. It renders HTML/CSS like a browser and outputs professional PDFs with excellent typography. If you know HTML and CSS, you already know how to use WeasyPrint.

Why Use WeasyPrint?

- **Modern Web Standards** - HTML5 + CSS3 support
- **Excellent Typography** - Uses HarfBuzz for text shaping
- **Familiar Syntax** - If you know CSS, you're done
- **Automatic Pagination** - CSS Paged Media support
- **Print-Ready** - Professional typesetting out of the box
- **Template Engines** - Works with Jinja2, Django templates, etc.
- **Preview in Browser** - Debug HTML before PDF generation
- **CSS Grid/Flexbox** - Modern layout capabilities

How It Works

WeasyPrint takes HTML content (string or file) and CSS styling, renders it like a browser would, and converts the rendered output to PDF. It supports CSS Paged Media for print-specific features like page breaks, headers, footers, and page numbering.

Basic Example:

```
from weasyprint import HTML

html_content = """
<html>
<head>
  <style>
    @page { size: letter; margin: 1in; }
    body { font-family: serif; }
    h1 { color: #0d47a1; }
  </style>
</head>
<body>
  <h1>My Document</h1>
  <p>Content here...</p>
</body>
</html>
"""

HTML(string=html_content).write_pdf("output.pdf")
```

Best Use Cases

 **Web Content to PDF** - Convert HTML pages to PDF

 **Markdown to PDF** - Markdown → HTML → PDF pipeline

 **Template-Based Documents** - Jinja2 + WeasyPrint for dynamic content

 **Web Scraping to PDF** - Scrape web pages and convert to PDF

 **Reports from HTML** - Generate reports using HTML/CSS

 **Rapid Prototyping** - Quick PDF generation with familiar web tech

Limitations & Considerations

- **System Dependencies** - Requires Cairo, Pango (may need installation)
- **Larger File Sizes** - PDFs tend to be larger than ReportLab
- **Slower Generation** - HTML rendering adds overhead
- **CSS Print Quirks** - Some CSS features behave differently in print
- **Less PDF Control** - Less control over PDF internals

Installation

```
pip install weasyprint
```

Note: May require system dependencies (Cairo, Pango, GDK-PixBuf) depending on your platform. Check WeasyPrint documentation for platform-specific installation instructions.

When to Choose WeasyPrint

Choose WeasyPrint if you're comfortable with HTML/CSS, have HTML/Markdown content, want rapid development, need web-to-PDF conversion, or are building template-based systems. It's perfect for teams with web development backgrounds.